



Sparse Interpretability of Audio Token Representations in Spoken Language Models

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Overview

Analyze and interpret frame-level audio token representations by training a Sparse Autoencoder (SAE) on encoder outputs of Qwen2-Audio.

Interpret latent features in SAE:

- Find that the audio encoder captures rich phonetic structure but offers limited word-level semantic abstraction.

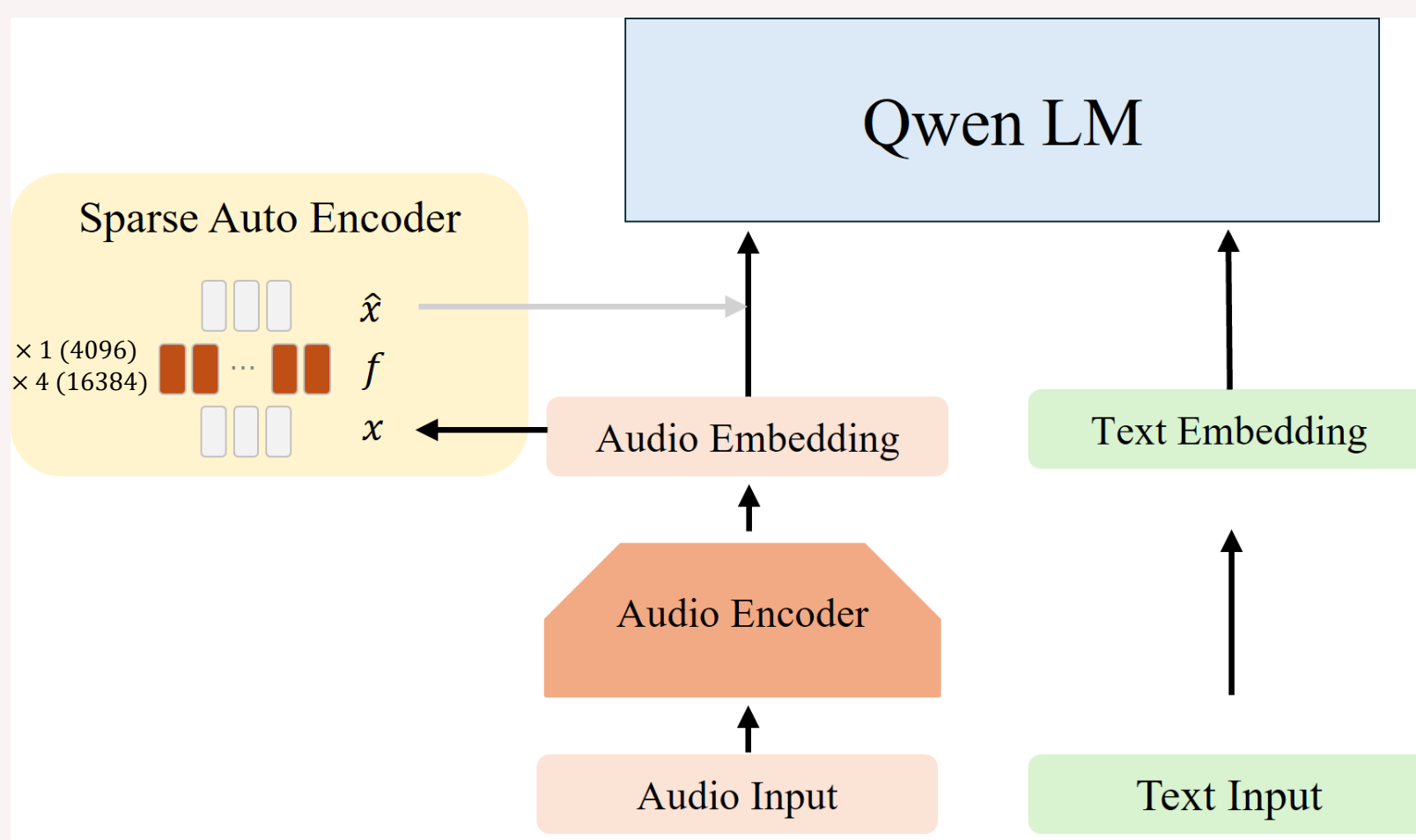
Motivation

- Whether **audio representations** in Qwen2-Audio serve as semantically aligned inputs as **text tokens** in language modeling.
- Sparse Autoencoder is a powerful **dictionary learning** method that distangle monosemantic features, resolving **superposition** structure within token space.

$$x^j \approx b + \sum_i f_i(x^j) \cdot d_i$$

Method

1. The Sparse Auto Encoder Framework



$$\begin{aligned} \bar{x} &= x - b_d && \text{Pre-encoder bias} \\ f &= \text{ReLU}(W_e \bar{x} + b_e) && \text{Sparse Activation} \\ \hat{x} &= W_d f + b_d && \text{Reconstruction} \end{aligned}$$

$$Loss = \frac{1}{N} \sum_{i=1}^N ||x^{(i)} - \hat{x}^{(i)}||_2^2 + \lambda ||f^{(i)}||_1$$

- Reconstruction loss + L1 sparsity loss

2. How to Train a Good SAE

- Improve Sparsity:** Top-k activation
- Avoid Inactivation** of latent features: Resampling Dead Neurons
- Dataset: 2 million audio tokens from Librispeech

3. Forced alignment with MFA

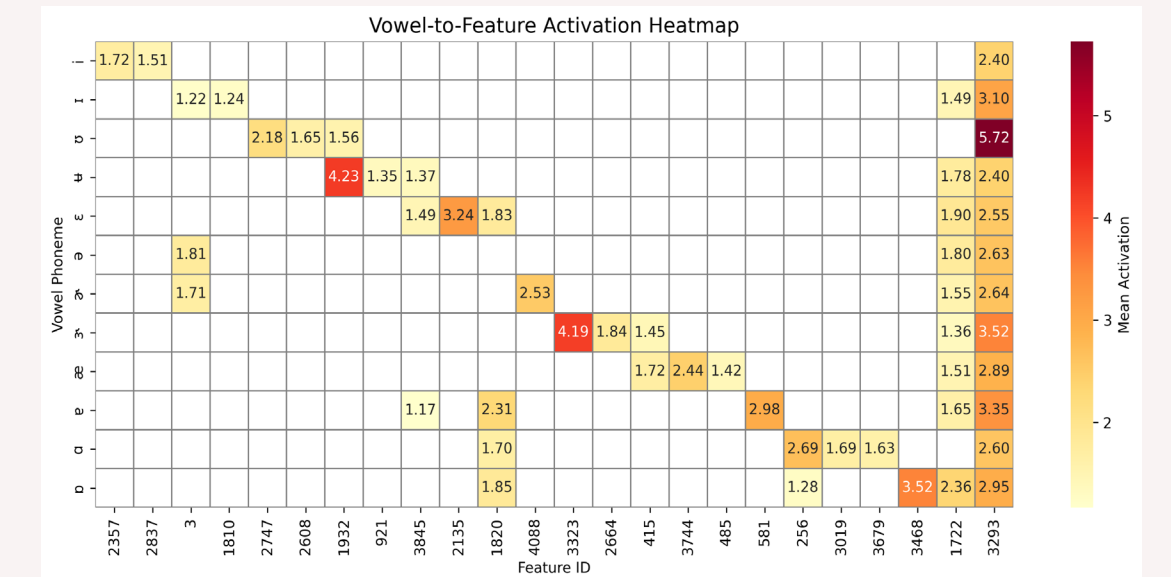
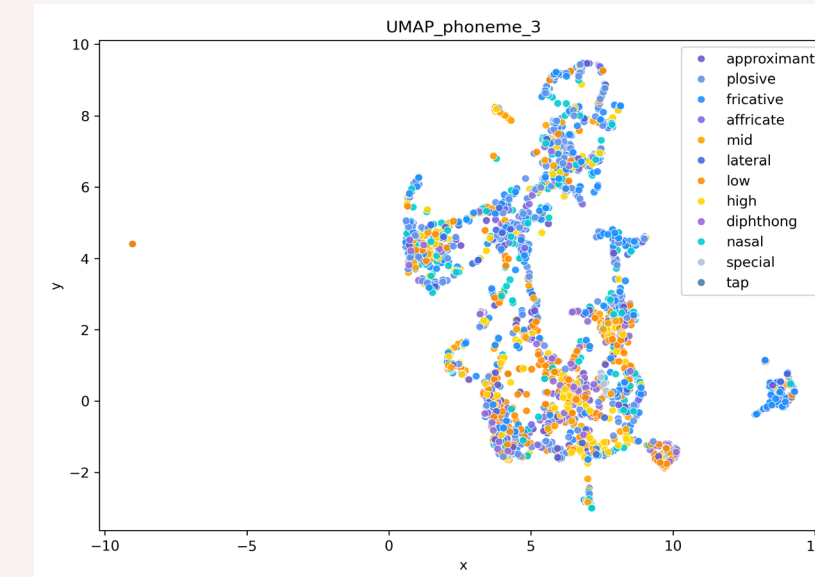
phoneme and word labels aligned to audio frames.

Analysis

Features: latent neurons of Sparse Autoencoder

1. Interpretability Analysis

■ Phonetic-level meaning



- separable between vowels and consonants, performs mono-semanticity.

Feature ID	Phoneme	Category	Example Word(s)
2570	/t/ (word-final)	Plosive	but, at
2990	/θ/ (word-initial)	Voiceless plosive	these
3293	/i/	Short front vowel	his, him
3055	/i:/	Long vowel	he, she
1722	/l/ (dark L)	Velarized consonant	hole, rule

- Context Sensitivity:** Same phoneme activates different features depending on its position within the word.

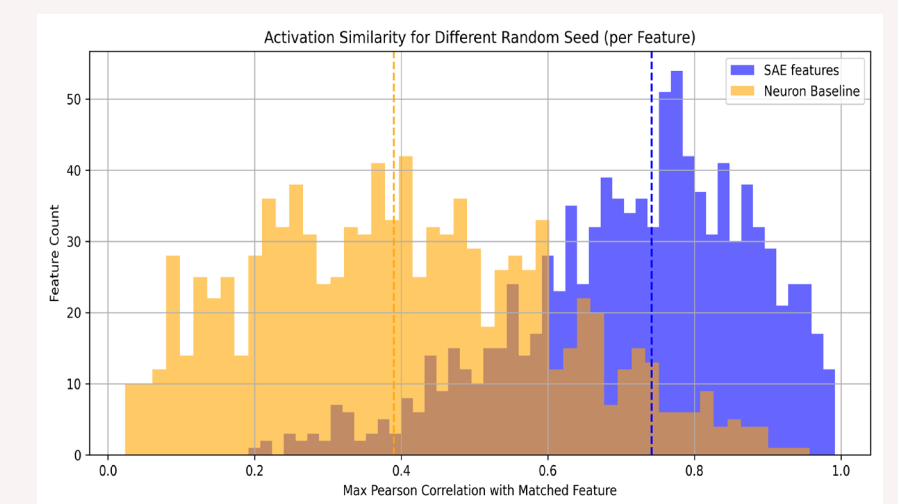
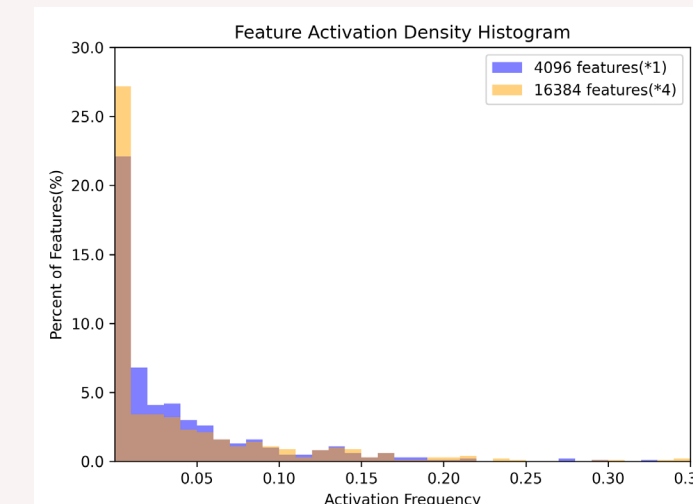
■ Word-level meaning (limited)

Feature	Word Meaning	Example words	Phoneme	Acoustic pattern
3293	Pronouns, Prepositions	his, him, her, they	/i/ /ə/	Vowel (short, weak)
2621	auxiliary verbs, modal verbs	had, would, could	/d/ /t/	Stop consonant
3673	Prepositions, Function words	to, for, of, at, the	/ə/, /æ/	schwa-like neutral vowels + soft consonant

- The “likely” word features all have underlying acoustic patterns

2. Robustness Analysis

Activation Similarity: $\text{Median}_{i=1}^d (\max_{j=1}^d \text{corr}(f_i^A, f_j^B))$



- Robust to different **dimension of features**, **random seed**

Conclusion

- Audio encoder captures rich phonetic structure. Some phoneme features capture position or context-sensitive patterns.
- While certain features show word-level patterns—often activating function words—their behavior is driven by acoustic regularities, not lexical identity. No evidence of linear representations is observed.

Future work

- Compare with other layerwise interpretability methods** (e.g., CCA, mutual information, probing, intervention)
- Extend analysis to the language model** (e.g., attention heads, unembedding layers)
- Examine richer audio features** (e.g., emotion, prosody, intonation, background noise, music)
- Apply sparse codes to downstream tasks** (discriminative and generative use cases)